

■ Regression Results — ELI5 Interpretation

Term	Meaning	Plain-English Explanation
Intercept = 0.722	Baseline tip when all predictors are zero.	If someone ordered a \$0 meal (theoretical), the model still “expects” about a \$0.72 tip baseline.
bill_total_usd = 0.094	Tip change per dollar of bill.	For every \$1 increase in the bill, the tip rises by roughly \$0.09. So a \$10 bigger bill adds about \$0.94 to the tip.
party_size = 0.180	Tip change per extra person.	Each additional person in the group increases the tip by about \$0.18, on average.
gender_Male = -0.027	Adjustment for male servers (vs. female).	Male servers earn around \$0.03 less per tip, holding other factors constant.
is_smoker_Yes = -0.084	Adjustment for smoking tables.	Smoking tables tip roughly \$0.08 less on average than non-smoking tables.
R ² = 0.469	Goodness of fit.	The model explains about 47% of the variation in tips — a decent but not perfect predictor.

■ ELI5 Summary

Imagine you’re guessing someone’s tip using clues: bill size, group size, smoking, and server gender. Each clue nudges your guess a little up or down.

Bigger bills → higher tips. Bigger parties → higher tips. Smokers and male servers → slightly lower tips. Your guessing game is right about **half the time (47%)**, so it’s helpful but not perfect.